

Homework 1

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August 29, 2026

1 Problem 1

Transcribed directly from my notebook:

$$\begin{aligned}y'(x) &= e^{y-2x} + 2 && \text{let } u = e^{-y}; \quad u' = -e^{-y}y' \\e^{-y}y' &= e^{-y}e^ye^{-2x} + 2e^{-y} \\-u' &= e^{-2x} + 2u \longrightarrow u' + 2u = -e^{-2x} \\\mu &= e^{\int 2 dx} = e^{2x} \\e^{2x}u' + 2e^{2x}u &= -e^{2x}e^{-2x} = -1 \\e^{2x}u &= -x + c \\e^{-y} &= e^{-2x}(-x + c) \\-y &= \ln |e^{-2x}(-x + c)| \\y &= 2x - \ln | -x + c |\end{aligned}$$

At $(0, -\ln 2)$,

$$-\ln c = -\ln 2 \longrightarrow c = 2$$

so

$$\therefore \quad y = 2x - \ln |2 - x| \quad \text{and} \quad y(1) = 2.$$

2 Problem 2

For each integrator, increasing the number of steps decreases the error because each individual step covers a smaller interval. It's extremely analogous to summation becoming smoother as your dx approaches 0.

RK2 and RK4 are improvements upon Euler step because they peek ahead at the function's behavior in order to more accurately choose an average slope to step through. RK4 uses more points in order to find that average slope, and is thus more accurate than RK2, which only samples the midpoint.

This is reflected in the numerical results. Euler had the largest RMS error for every step count, while RK2 converged much more quickly. RK4 reached errors near machine precision for the largest step counts, so increasing the number of steps further did not provide a meaningful improvement.

See problem2.ipynb or problem2.pdf.

3 Problem 3

RK45 greatly improved the efficiency by only emphasizing detail where it was needed. As the relative tolerance was decreased, the RMS error generally decreased while the number of function evaluations increased.

At the strictest relative tolerance tested, RK45 reached an RMS error on the order of 10^{-12} with 196 function evaluations. This shows the advantage of adaptive stepping, since high accuracy can be reached without uniformly using an extremely small step size throughout the entire interval.

The error did not decrease perfectly monotonically with tolerance because the analytical solution crosses zero within the interval. Since the assigned error is a relative error, values near this zero crossing can become artificially large depending on the locations of the adaptive support points.

See problem3.ipynb or problem3.pdf.

4 Problem 4

For each body i , the equations of motion can be written as

$$\dot{x}_i = v_{x,i}, \quad \dot{y}_i = v_{y,i}, \quad \dot{z}_i = v_{z,i}$$

and

$$\begin{aligned} \dot{v}_{x,i} &= G \sum_{j \neq i} m_j \frac{x_j - x_i}{r_{ij}^3}, \\ \dot{v}_{y,i} &= G \sum_{j \neq i} m_j \frac{y_j - y_i}{r_{ij}^3}, \\ \dot{v}_{z,i} &= G \sum_{j \neq i} m_j \frac{z_j - z_i}{r_{ij}^3}. \end{aligned}$$

where

$$r_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2}.$$

For the normalized units used here, $G = 1$.

The adaptive RK45 integration produced 7042 support points while evolving the Solar System for 100 years. The resulting trajectories remained approximately stable, although some accumulated numerical error became visible over the long integration.

The total energy was also calculated at every timestep as a check on the numerical solution. The maximum relative change in total energy over the 100 year simulation was approximately

$$6.38 \times 10^{-4},$$

or about 0.064%. This small drift is consistent with accumulated numerical integration error over the simulation.

See `problem4.ipynb` or `problem4.pdf`, and `problem4.mp4`.